LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Warning lights, caution lights, indicating or advisory lights, magnetic indicators, central warning systems, electronic display systems
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Warning lights, caution lights, indicating or advisory lights, magnetic indicators, central warning systems, electronic display systems.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Central Warning System panel operation; Safety: Do not touch live circuits; Supervision: Instructor monitors all interactions.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a warning light in an aircraft?
(b) How does a caution light differ from an advisory light?
(c) What role do magnetic indicators play in flight instrumentation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Difference between warning and caution lights • Function of indicating and advisory lights • Operation of magnetic indicators • Overview of central warning systems • Introduction to electronic display systems	L & Ds	PPT, IB	15 min
(b)	• Central warning system architecture • Electronic display system components • Integration of warning lights with generators • Actuator control via warning signals • Engine starting system light indicators	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of warning panel • Identification of light colors and meanings • Troubleshooting central warning faults • Review of electronic display interfaces • Correlation with circuit protection devices	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the sequence of events in a central warning system activation.
(b) How do electronic display systems replace traditional magnetic indicators?

12. Home Assignment and Follow Up	Prepare a brief report on the cleaning procedures for electrical connectors found in electronic display systems, referencing the need for heavy-duty cleaners as covered in LP 24.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need for cleaning the electrical connectors Use of heavy-duty cleaner, cleaning fluid for electrical, electronics and allied components. Safety precautions & Practical on cleaning of connecters, terminals.	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min